

Introduction to causal inference
using machine learning methods

Chapter 0:
Overview of the course

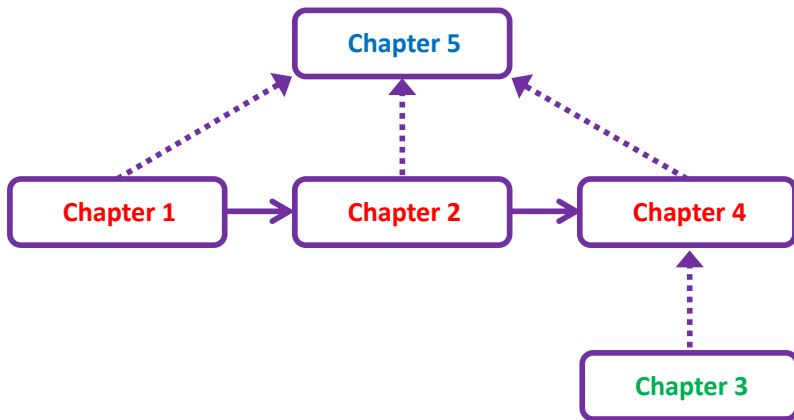
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Organization of this course

- 1 The basics of causal inference
- 2 Identification and estimation of an average treatment effect
- 3 Super learning
- 4 Efficient, doubly-robust estimation of an average treatment effect
- 5 Additional topics (time permitting)

Organization of this course



LEARNING OBJECTIVES:

At the end of this short course, we expect each student to have an understanding of:

- 1 the role of a causal model in making explicit the available background knowledge;
- 2 the definition of counterfactuals and their role in defining causal effects that address the scientific question of interest;
- 3 the key identification formulas for linking counterfactual and observed data parameters in the context of observational data;
- 4 the various traditional techniques for causal inference and their limitations;
- 5 the benefits of the super learner as an optimal, data-driven and pre-specified approach to flexible estimation;
- 6 the benefits of more advanced techniques for causal inference, including the augmented IPTW and TMLE estimators;

Organization of this course

A few guidelines for the day...

- Audience is heterogeneous — we will try to cater to everyone as best we can.
- Slides include more details than strictly needed to understand the material.
- Please feel absolutely free to ask clarification questions at any time, but reserve 'enrichment' questions for the online chat or for the end of the session.